

Prostate Health Index and Early Cancer Biomarker Panel

Multi-Analyte Immunoassay Panel for Prostate Cancer Risk Stratification (Synthetic Demonstration)

Patient Demographics	Ordering Clinician	Specimen Chain of Custody
Name: Arthur B. Sample DOB: 02/14/1959 (Age 67) Sex: Male Patient ID: SYN-MRN-3341890	Provider: Dr. David Reynolds, MD Facility: Urologic Specialty Clinic NPI: 9092837465 Location: Minneapolis, MN	Specimen ID: SYN-SPEC-PHI-8821 Specimen Type: Serum Tube: SST Gold Top (5.0 mL) Collection Date: Aug 10, 2026 Received Date: Aug 11, 2026 Report Date: Aug 12, 2026

Result: Elevated Risk (High Risk for Prostate Cancer)

Status: Final
Index (phi): 47.8

Test Result Summary: Patient has an elevated Prostate Health Index (phi = 47.8) in this synthetic test result, indicating an elevated probability (36.0% - 55.0%) of clinically significant prostate cancer (Gleason Score >= 7) on biopsy.

Quantitative Immunoassay Biomarkers

Biomarker / Assay	LOINC	Result	Units	Reference Range	Flag
Total PSA	2857-1	5.6	ng/mL	0.0 - 4.0	High (H)
Free PSA	10886-0	0.58	ng/mL	N/A	Normal
% Free PSA	19195-7	10.4	%	> 25.0 %	Low (L)
[-2]proPSA	72304-9	19.2	pg/mL	N/A	Normal
Prostate Health Index (phi)	72305-6	47.8	{score}	< 27.0 (Low Risk)	High (H)

Clinical Interpretation & Recommended Next Steps

The Prostate Health Index is calculated via the formula: $\text{phi} = ([-2]\text{proPSA} / \text{Free PSA}) * \sqrt{\text{Total PSA}}$. A phi value ≥ 36.0 is associated with a 36.0% to 55.0% probability of cancer on biopsy. In men aged 50 and older with total PSA in the 4.0 to 10.0 ng/mL diagnostic gray zone, this elevated score provides clinical justification for mpMRI and targeted prostate biopsy.

- **Multiparametric prostate MRI (mpMRI):** 3-Tesla pelvic mpMRI with PI-RADS scoring.

- **Urology consult:** Referral for MRI-fusion targeted and systematic transperineal or transrectal prostate biopsy.

Methodology & Limitations

Methodology: Total PSA, Free PSA, and [-2]proPSA are quantitatively measured in serum using automated chemiluminescent immunoassay analyzers calibrated against WHO standards. The phi score is derived mathematically.

Limitations: phi results should be evaluated in conjunction with digital rectal examination (DRE), prostate volume, family history, and clinical risk factors.

Laboratory Director Authorization: Dr. Keith Carlson, MD, FCAP | Electronically Signed | Date: Aug 12, 2026 | CLIA ID: 00D1122334